

EEET202

CHINES & TRANSFORMERS

MODULE:3

re:5 – Braking of DC Motors

of operation of DC motors and select appropriate motor types for

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Syllabus

generation of torque, torque equation, performance characteristics – starters – 3 point and 4 point starters (principle only).

tors - field control, armature control. Braking of dc motors. Power and efficiency. Testing of dc motors - Swinburne's test, Hopkinson's test. DC motor applications – numerical problems.

Electric Braking of DC Motor

Stopping but stopping

Electrically or mechanically

Braking - stopping the motor by some means without any mechanical work

Braking motor works as generator and produces a negative torque which opposes the

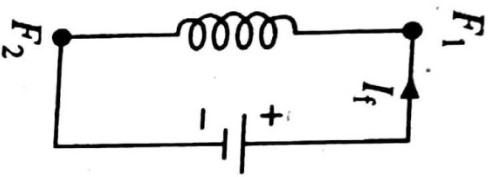
direction of electric braking

Braking

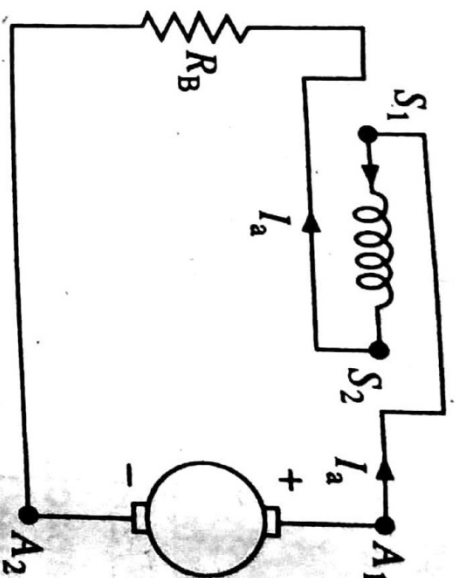
Electric Braking

Dynamic Braking

are disconnected from the source and
dissipate energy in R_B and R_a



excited motor

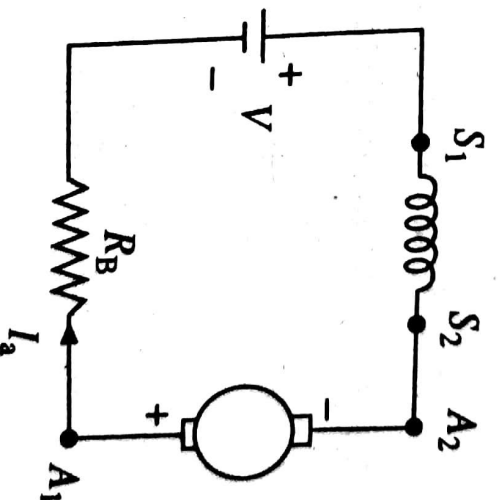
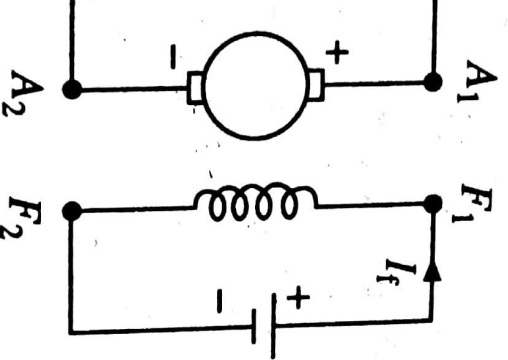


(b) Series motor

Plugging

the motor is reversed so that it assists the back armature current in reverse direction. carries motor, armature alone is reversed. t braking.

at zero at zero speed, when used for stopping a be disconnected when close to zero speed.



(b) Series

arately excited

Regenerative Braking

Energy is supplied back to the source.

When following condition should be satisfied.

Armature voltage should be greater than supply voltage

Motor should be run at above rated speed

As speed increases, armature current and torque decreases. Thus regenerative braking is not